

Amplitude-dependent response of a CH₄/Air swirl-stabilized flame to equivalence ratio oscillations

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ABSTRACT

Thermoacoustic instability is a critical issue in gas-turbine and aero-engine systems, typically involving coupled velocity and equivalence ratio (ϕ) oscillations. Under the assumption that velocity and ϕ oscillations do not interact, a multi-input single-output (MISO) model in which the flame transfer function is replaced by the flame describing function can be applied to such complex systems in the nonlinear regimes, but sufficient experimental validation is lacking. This study systematically examines the response of a premixed CH₄/air swirl-stabilized turbulent flame to ϕ oscillations of various amplitudes at a fixed modulation frequency of 10 Hz using a dual-valve modulation system. And then, appropriate operating cases were chosen to validate the MISO model. As mean equivalence ratios (ϕ_0) equal to 0.65 or 0.9, the flame enters a pronounced nonlinear regime when amplitude exceeds 0.02, necessitating a third-order Taylor expansion within the quasi-steady state assumption (QSSA) framework for precise modeling. As $\phi_0 = 1.1$, the gain of flame describing function is largely independent of the amplitude and can be accurately predicted by the QSSA framework at low amplitudes ($|\hat{\phi}|^{\text{ref}} < 0.1$). However, when ϕ oscillations encompass the steady-state curve's inflection point and even enter the regime of abrupt flame morphological changes, the QSSA predictions show significantly larger deviations. After confirming that velocity and ϕ oscillations are decoupled, the cases with $\phi_0 = 0.9$ were used to validate the MISO model combining with a single-valve modulation system. And the results show that the MISO model accurately captures the amplitude dependence of the flame describing function even in the nonlinear regimes. The study further indicates that the relative deviations introduced by the application of the MISO model arise from variations in the instantaneous velocity. While the instantaneous velocity alters the overall flame morphology and consequently affects the spatial distribution of CH* chemiluminescence.

Novelty and significance statement

This study first presents the quantitative assessment of relative deviations introduced by the application of the multi-input single-output (MISO) method in nonlinear regimes. When velocity and equivalence ratio (ϕ) oscillations are decoupled, experiments were performed using a single-valve system (combined velocity and ϕ oscillations) and a dual-valve system (ϕ oscillation) adapted from our previous work. During these experiments, the response of a swirl-stabilized flame to low-frequency (10 Hz) ϕ oscillations was quantified, and the influences of mean equivalence ratio and amplitude were identified. The quasi-steady-state analysis (QSSA), derived from the highest-order Taylor expansion, was used to predict the amplitude dependence of the describing function. And the ranges in which the QSSA yields accurate predictions and where it fails were also reported.

1. Introduction

High-efficiency and low-emission combustion have become key development goals for next-generation gas turbines. Lean premixed combustion technology has emerged as a promising approach to meet these requirements. However, lean premixed operation is inherently more susceptible to thermoacoustic instabilities that can adversely affect

the performance and structural integrity of gas turbines and aero-engines [1,2]. It is well established that the driving mechanisms of thermoacoustic instabilities can be broadly categorized into two primary types: velocity perturbations and equivalence ratio oscillations [3, 4].

Extensive research has been conducted on the flame response to velocity perturbations [5–8], whereas studies focusing on the response

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to equivalence ratio oscillations remain relatively limited [9–12]. A key challenge is that equivalence ratio oscillations are difficult to modulate and frequently couple with velocity fluctuations [13]. Especially for real gas turbine combustors, combustion typically initiates before the fuel and oxidizer are perfectly premixed and homogenized. And the resulting flame can be therefore classified as non-premixed flame [14–16] or partially premixed flame [17–19]. When non-premixed flames or partially premixed flames are subjected to acoustic perturbations, the resulting out-of-phase (non-synchronous) response between the fuel injector [20] and the air inlet strongly promotes equivalence ratio oscillations [4,21,22], which further complicates the overall unstable combustion process. To address this, Polifke et al. [23–25] proposed the multi-input single-output (MISO) model to extract the describing functions of the flame response to equivalence ratio oscillations in the presence of velocity oscillations, as shown in Eq. (1):

$$\frac{\hat{I}}{\bar{I}} = G \frac{\hat{u}}{\bar{u}} + F \frac{\hat{\phi}}{\bar{\phi}} \quad (1)$$

where $\hat{I}$ and $\bar{I}$ denote the frequency-domain representation and the mean value of a parameter, respectively. I represents the global chemiluminescence intensity of the flame, while u and ϕ represent the velocity and equivalence ratio measured at the reference point. G and F are the transfer functions between chemiluminescence intensity and velocity or equivalence ratio, respectively. In the framework of the MISO model, the transfer functions associated with the velocity oscillation alone, as well as those for the coupled velocity and equivalence ratio oscillations can be obtained separately. Therefore, the transfer function associated with pure equivalence ratio oscillations can be evaluated according to Eq. (1). This approach is straightforward to implement and particularly suitable for complex flame systems. Since the MISO model is restricted to linear regimes, it must be rigorously tested whether transfer functions can be substituted with describing functions (FDF) under the decoupling assumption (velocity and equivalence ratio oscillations are non-interfering) [26].

In our previous work on laminar flame [27], a custom developed dual-valve equivalence ratio modulation system was presented, and the FDF between heat release rate and pure equivalence ratio oscillations

was analyzed. A small perturbation model and a quasi-steady state assumption (QSSA) mathematical model were employed to explain the limited frequency and amplitude dependence of FDF of laminar premixed flames. In present work, the aim is to assess the MISO model in the nonlinear response regime of swirl-stabilized turbulent flames subjected to both velocity modulation as well as equivalence ratio modulation of varying amplitudes at a fixed modulation frequency of 10 Hz. Controlled modulation of velocity and equivalence ratio oscillations is achieved by employing both single-valve and dual-valve modulation systems. Then, the MISO model is employed to derive the FDF for equivalence ratio oscillation alone based on the data from single-valve modulation system. This FDF is then compared with the one obtained directly from the dual-valve system, allowing quantitative assessment of the relative deviations introduced by using the MISO model in nonlinear regimes. Furthermore, during the construction of FDF, the applicability of the QSSA mathematical model is also evaluated, which was previously not adequately addressed due to the limited range of modulation amplitudes.

2. Experimental setup

The structural schematic of the central bluff-body stabilized swirl burner is shown in Fig. 1(a-b). The exit diameter is 25.4 mm, and the bluff-body diameter is 6 mm. The top of the bluff body is hemispherical, with its apex located approximately 5 mm upstream of the exit. The schematic of the swirler is illustrated in Fig. 1(c). The swirl number (S , equal to 0.43) is defined by Eq. (2) [28]:

$$S = \frac{2}{3} \left[\frac{1 - (D_h/D_n)^3}{1 - (D_h/D_n)^2} \right] \tan \eta \quad (2)$$

where D_h and D_n are the inner and outer diameters, respectively. η is the vane angle relative to the centerline.

Upstream of the swirler is a straight tube with a length of 230 mm (in which $Re=7600$), connected to the equivalence ratio modulation system, as illustrated in Fig. 1(d). The modulation system is interchangeable and can be operated using either a dual-valve or a single-valve configuration. A detailed description of the dual-valve system is provided in [27]. In brief, the equivalence ratio modulation system features

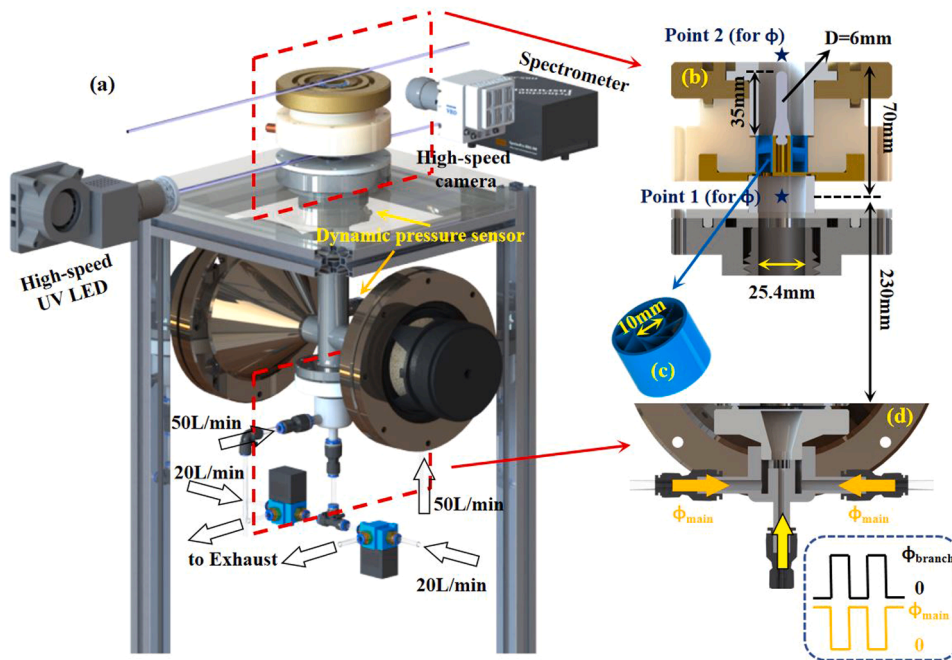


Fig. 1. (a) Experimental setup for a low-swirl flame with equivalence ratio modulation. (b) Cross-sectional view near the outlet of the swirl burner. (c) Schematic of the swirler. (d) Scheme of the valve time gating for equivalence ratio modulation.

two inlets, referred to as the "main channel" and the "modulation channel." The main channel is located on both sides of the burner, operating at a flow rate of 110 L/min and a corresponding equivalence ratio of ϕ_{main} . Then, the mixture enters the burner through circumferentially distributed small holes. The modulation channel delivers a flow rate of 20 L/min, which is injected through the bottom inlet of the burner. Once the two gas streams have mixed, they enter the straight tube, pass through the swirler, and then ignited downstream of the bluff body, forming a typical "V"-shaped swirl flame (shown in Fig. 2(a)).

For the dual-valve system, the modulation channel is connected via a three-way junction to two solenoid valves, defined as the "primary valve" and the "secondary valve." The mixture with an equivalence ratio equal to ϕ_{branch} flows through the primary valve with a duty cycle of 50%. When the primary valve is closed, the secondary valve (ϕ_{main}) opens, operating with a 50% duty cycle. Alternating the operation of the two valves ensures that the flow rate in the modulation channel remains constant, thereby generating periodic equivalence ratio oscillations without inducing any velocity oscillations. When the system is operated under the single-valve modulation configuration, the mean flow remains the same as in the dual-valve configuration, but the modulation channel is connected with only one valve. The action of the single valve introduces parasitic velocity oscillations along with the equivalence ratio modulation.

Fig. 2 illustrates the response of the "V"-shaped swirl flame to the two modulation schemes at a zero-oscillation amplitude. Under the single-valve modulation, the flame morphology varies significantly over time. Approximately 30 ms after the valve opening at t_0 , the flame develops pronounced wrinkles, and its height is notably lower than that observed at 80 ms. In contrast, under dual-valve modulation, the flame morphology remains nearly unchanged at different time. Velocity fluctuations upstream of the swirler at point 1 measured using a hot-wire system (HWA, Streamline Pro, 55P11) were approximately 0.4% and 9.8% for the dual-valve system and the single-valve system, respectively. For the equivalence ratio calibration, trace amounts of SO_2 gas were seeded into the modulation flow, and equivalence ratio oscillation was measured 10 mm upstream of the swirler (cf. point 1 in Fig. 1(b)) using SO_2 broadband absorption spectroscopy (1 kHz), as detailed in a previous study [27]. Accordingly, the nominal mean equivalence ratio (ϕ_0 , defined as the arithmetic average of the maximum and minimum values, ϕ_{min} and ϕ_{max} , within a modulation period) and the modulation amplitude ($|\hat{\phi}|^{\text{ref}} = (\phi_{\text{max}} - \phi_{\text{min}})/2$) were estimated. In the present work, the equivalence ratio modulation frequency (f) was fixed at 10 Hz, and

conditions with $\phi_0 = 0.65, 0.9$ and 1.1 , with varying $|\hat{\phi}|^{\text{ref}}$ up to 0.4 were investigated.

The selection of the mean equivalence ratios is based on the steady-state relationship between the chemiluminescence intensity and equivalence ratio shown in Fig. 3(a). A mean equivalence ratio of 0.65 was chosen because the flame operates near the extinction limit under this condition. A value of 0.9 was selected because the chemiluminescence intensity varies monotonically with the equivalence ratio within the operable range of equivalence ratio oscillations. Finally, $\phi_0 = 1.1$ was selected to investigate more complex flame response phenomena: as the amplitude of equivalence ratio oscillation increases, CH^* within the oscillation range evolves from a monotonic dependence to a non-monotonic one, and ultimately becomes discontinuous, as shown in Fig. 3(a-b).

Using the absorption spectroscopy system described above, 100 repeated measurements (each with a duration of 1 s) were performed to estimate the uncertainties in the mean equivalence ratio $\bar{\phi}$ and modulation amplitudes $|\hat{\phi}|$ at Points 1 and 2, as illustrated in Fig. 3(c-d). At Point 1, the uncertainties were approximately 3% and 12% for the dual-valve modulation system, and 1% and 7% for the single-valve system, respectively. Compared with Point 1, the uncertainties at Point 2 may be

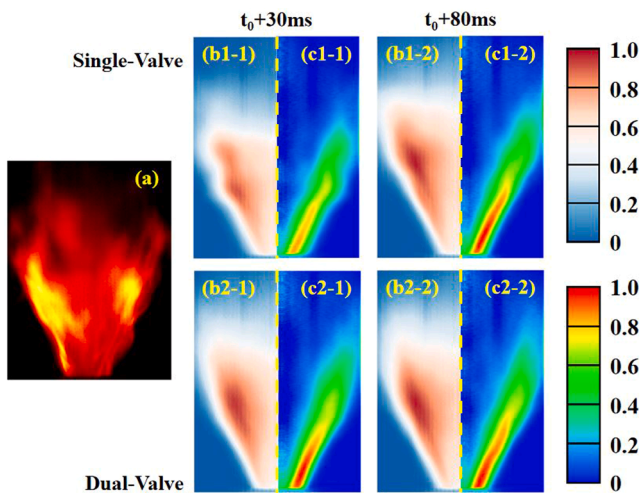


Fig. 2. (a) Instantaneous image of the unmodulated flame. Phase-locked averaged flame images (left images) at the selected time instants under (b) single-valve or (c) dual-valve modulation, together with the corresponding Abel inversion results (right images).

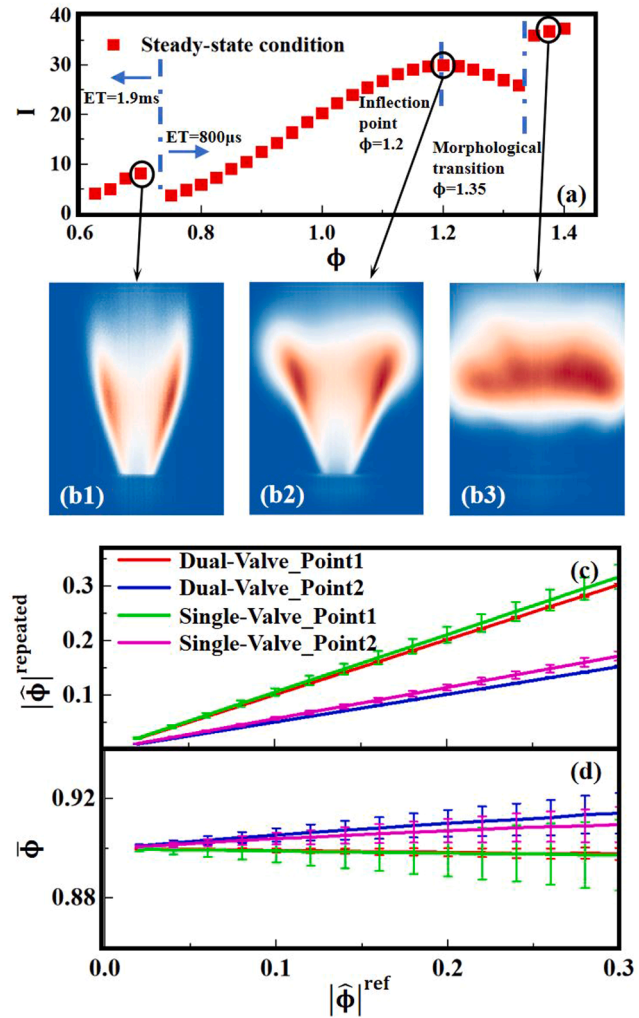


Fig. 3. (a) Relationship between the equivalence ratio and the globally integrated CH^* chemiluminescence intensity for the unmodulated flame. (b) Time-averaged CH^* distributions without modulation at $\phi = 0.7, 1.2$, and 1.375 . Variations in (c) the repeatedly measured modulation amplitudes and (d) the time-averaged equivalence ratio at Points 1 and 2 as a function of the modulation amplitude at the reference point (Point 1).

affected not only from the testing system, but also from the influence of surrounding fresh air and the central recirculation zone. Although, the uncertainties exhibited a clear amplitude dependence (increasing with the modulation amplitude), they exerted a negligible influence on the analyses of the amplitude- and mean equivalence ratio-dependent behavior.

The flame dynamics were captured by a high-speed camera (PHANTOM, operating at 500 Hz) equipped with a band-pass filter (430 ± 5 nm for the detection of CH^* emission). As shown in Fig. 3(a), the exposure time (ET) was adjusted to maintain adequate signal-to-noise levels. Specifically, a longer exposure of 1.9 ms was employed for equivalence ratios below 0.725, whereas an exposure time of 0.8 ms was used for the remaining conditions. Although the validity of CH^* as a proxy for heat release rate under varied conditions is contested, it remains one of the most accessible experimental measures of flame response. This study focuses on validating the QSSA and MISO models, which do not require heat release rate oscillations. For MISO validation in Section 4.2, CH^* -based flame morphology is examined to explain deviations in the model's nonlinear regime.

3. Methodology

Polifke et al. [29] have established that, the gain of the flame transfer function could be fully constrained by global conservation of mass, energy, and momentum. And high-order Taylor expansion was used to predict the amplitude-dependence of the flame describing function in our previous work [27]. Based on these, the highest-order Taylor expansion at different ϕ_0 extracted from the steady-state curve in Fig. 3 (a) is employed to determine the describing function gain between CH^* intensity and equivalence ratio oscillations in the low-frequency limit (10 Hz), namely, quasi-steady state assumption (QSSA) given by Eqs. (3–4).

$$\frac{\Delta I/I_0}{\Delta \phi/\phi_0} = \frac{\phi_0}{I_0} \sum_{n=1}^3 \frac{1}{n!} \frac{\partial^n I}{\partial \phi^n} (\Delta \phi)^{n-1} \quad (3)$$

$$|F|_{f \rightarrow 0} = \left(\left| \frac{\Delta I/I_0}{\Delta \phi/\phi_0} \right|_{\Delta \phi > 0} + \left| \frac{\Delta I/I_0}{\Delta \phi/\phi_0} \right|_{\Delta \phi < 0} \right) / 2 \quad (4)$$

As for the MISO model, it was originally derived under linear conditions. Then, Ćosić et al. [26] sought to extend the applicability of the MISO model to the nonlinear regime by replacing the linear transfer functions with describing functions, and assuming that the velocity and equivalence ratio oscillations are decoupled. In this paper, experimental investigations (with modulation frequency equal to 10 Hz) are conducted to verify this extension. At this frequency, the dissipative effects on the mixture composition are not significant. And the spatial wavelength of the equivalence ratio oscillation (approximately 0.5 m, larger than the length of flame (≈ 0.05 m)) is weakly affected by the small velocity oscillations ($<10\%$). This indicates that the equivalence ratio oscillation is largely decoupled from the velocity oscillation. This conclusion is supported by Fig. 3(c–d): regardless of whether single-valve modulation or dual-valve modulation is employed, the amplitudes of the equivalence ratio oscillation at Point 2 are almost constant once the equivalence ratio oscillation at Point 1 is controlled. In addition, the velocity oscillation is mainly induced by valve actuation, and it will not be affected by the presence of equivalence ratio oscillations.

Accordingly, for the single-valve modulation, controlled experiments with (1) pure velocity perturbations as well as (2) coupled velocity and equivalence ratio perturbations were considered. The velocity perturbations in the two experiments were kept identical. This allowed the describing function for pure velocity perturbation, G , and the normalized chemiluminescence intensity fluctuation, $\hat{I}/\bar{I}$ under coupled perturbations (Eq. (1)) to be determined separately. Therefore, the

describing function for pure equivalence ratio perturbations, F , can be estimated using Eq. (1), which is then compared with that measured directly from the dual-valve modulation system. The corresponding relative difference, ε , was then evaluated using Eq. (5),

$$\varepsilon = \frac{|F^s - F^d|}{|F^d|} \quad (5)$$

where the superscripts s and d denote the corresponding describing functions obtained from the single-valve and dual-valve systems, respectively.

To further investigate the origin of ε introduced by the MISO, the time-resolved image sequences were processed using a linear superposition method. Specifically, the images obtained at $|\hat{\phi}|^{\text{ref}} > 0$ were subtracted frame by frame by those acquired at $|\hat{\phi}|^{\text{ref}} = 0$. Phase-locked averaging was then applied to the instantaneous difference images. This procedure effectively retains the variations induced by equivalence ratio oscillations. Subsequently, these single-valve difference images were subtracted from the images obtained using the dual-valve modulation system, as expressed in Eq. (6), resulting in a 2D signal difference ξ at the respective phases.

$$\xi = \left(I_{|\hat{\phi}|^{\text{ref}} > 0}^s - I_{|\hat{\phi}|^{\text{ref}} = 0}^s \right) - \left(I_{|\hat{\phi}|^{\text{ref}} > 0}^d - I_{|\hat{\phi}|^{\text{ref}} = 0}^d \right) \quad (6)$$

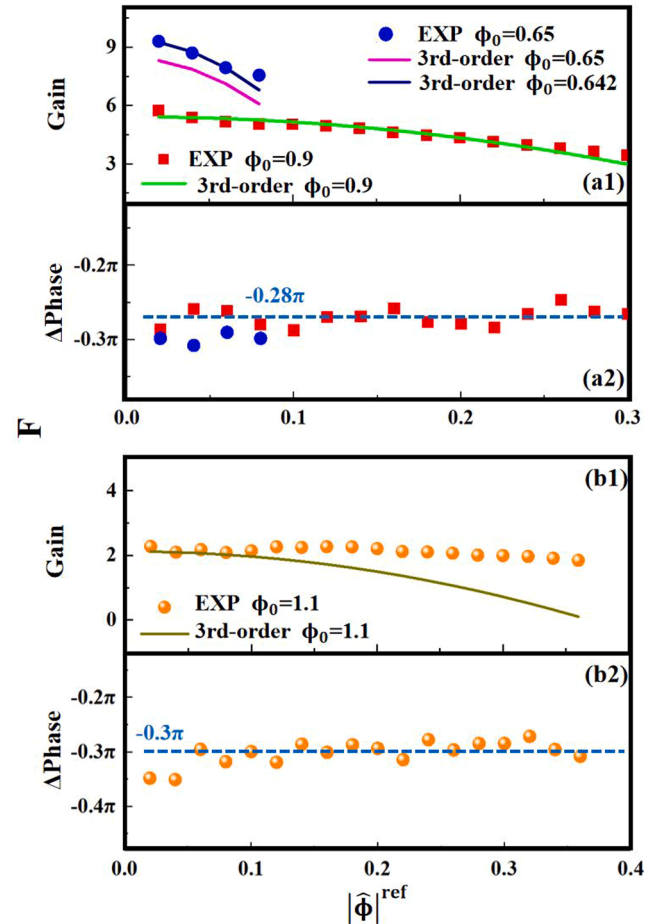


Fig. 4. The describing function F between the global CH^* chemiluminescence intensity and the equivalence ratio modulation at 10 Hz as a function of the nominal equivalence ratio modulation amplitude, $|\hat{\phi}|^{\text{ref}}$, for different nominal mean equivalence ratios (ϕ_0) of (a) 0.65 and 0.9 and (b) 1.1 at a modulation frequency of 10 Hz.

4. Results

4.1. Amplitude dependence

The describing function of the chemiluminescence intensity, F , obtained at $\phi_0 = 0.9$, is shown in Fig. 4(a). And the QSSA results using the third-order Taylor expansion are presented as a solid line. The gain of F gradually decreases with increasing amplitude, indicating that the flame enters the nonlinear regime when the $|\hat{\phi}|^{\text{ref}}$ exceeds 0.02. Moreover, the amplitude dependence of the gain can be accurately predicted using the QSSA when the third-order (highest-order) Taylor expansion is applied. The phase remains a constant value and is largely insensitive to changes in modulation amplitude. For $\phi_0 = 0.65$, the gain and phase exhibit an amplitude dependence similar to that observed at $\phi_0 = 0.9$. The best agreement with the experimental results is achieved under the QSSA when the third-order Taylor expansion and slightly shifted nominal ϕ_0 are applied. Considering the experimental uncertainty in ϕ_0 , the shift of ϕ_0 to 0.642 is acceptable.

For $\phi_0 = 1.1$, as shown in Fig. 4(b), the gain exhibits almost no amplitude dependence over a wide range of $|\hat{\phi}|^{\text{ref}}$. When the amplitude $|\hat{\phi}|^{\text{ref}}$ is below 0.1, the gain can be accurately predicted using the third-order Taylor expansion within the QSSA. As $|\hat{\phi}|^{\text{ref}}$ increases and the equivalence ratio oscillations span the inflection point of the steady-state function of $I(\phi)$ curve in Fig. 3(a), the predictive accuracy deteriorates. Then, as $|\hat{\phi}|^{\text{ref}}$ further increases, and the range of equivalence ratio oscillations extend into the region where the flame morphology changes. The luminosity variation can no longer be described as a continuous function, leading to progressively larger deviations in the QSSA predictions.

To further explain the decrease in the gain of the flame describing function with increasing $|\hat{\phi}|^{\text{ref}}$ under lean-combustion conditions, the condition with $\phi_0 = 0.9$ is examined as a representative case. The phase corresponding to the maximum CH^* intensity during the fluctuation of I is defined as 0° , while the phase corresponding to the minimum CH^* intensity is defined as 180° . Images at these two phases were extracted and phase-locked averaged. Subsequently, the phase-locked averaged images of the steady flame at $\phi = 0.9$ was subtracted from these time-averaged images to obtain ΔI . The results, normalized by the maximum intensity of the steady flame image, are shown in Fig. 5(a). The spatial distribution of ΔI indicates a globally consistent response under this condition. At the maximum brightness phase (0°), the intensity over the entire flame is higher than the mean field, whereas at the minimum phase (180°), it is uniformly lower.

The ΔI images were subsequently radially averaged to obtain the axial distribution of the normalized ΔI , as shown in Fig. 5(b). When $|\hat{\phi}|^{\text{ref}}$ is below 0.2, the absolute magnitude of ΔI increases markedly with the modulation amplitude at both the 0° and 180° phases. However, when the amplitude exceeds 0.2, the spatial distribution of ΔI at the 0° phase changes significantly: the root of the flame becomes almost insensitive to further increases in $|\hat{\phi}|^{\text{ref}}$, while the intensity in the upper region continues to increase. In contrast, at the 180° phase, the spatial distribution of ΔI remains largely unchanged, although the overall increase in its magnitude becomes substantially weaker. To further quantify this trend, the axial distributions at the two phases were spatially averaged to obtain the variation of the globally averaged ΔI with respect to $|\hat{\phi}|^{\text{ref}}$. The results show that ΔI at the 0° phase varies almost linearly with $|\hat{\phi}|^{\text{ref}}$, whereas the variation at the 180° phase becomes significantly weaker at larger amplitudes. Therefore, the reduced sensitivity of the chemiluminescence intensity to $|\hat{\phi}|^{\text{ref}}$ at the 180° phase is primarily responsible for the observed decrease in the gain of the flame describing function with increasing amplitude.

For $\phi_0 = 1.1$, as shown in Fig. 6(a), the global distribution of ΔI exhibits pronounced spatial heterogeneity. At the 0° phase, ΔI is negative in the flame root, whereas it is positive in the upper region. In both

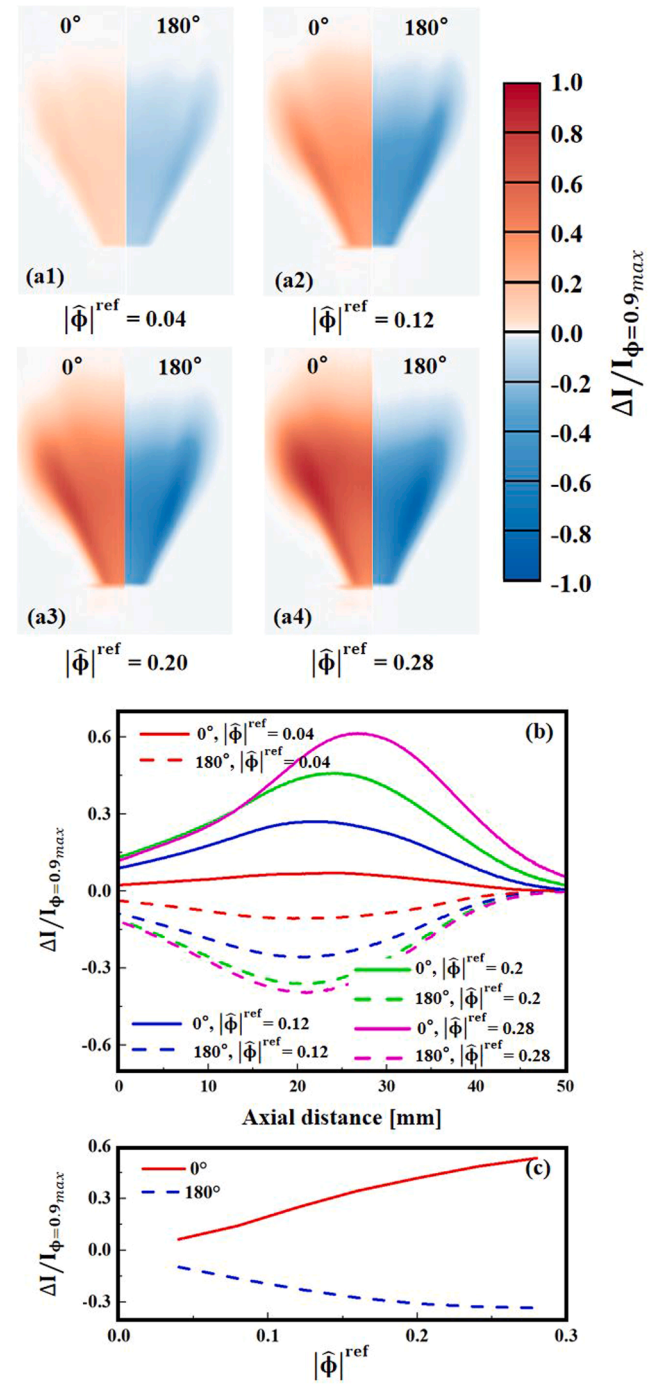


Fig. 5. (a) Phase-locked averaged CH^* chemiluminescence images relative to the mean image at 0° and 180° phases, with their (b) radially averaged intensity along the axial direction, and (c) spatially averaged intensity for $|\hat{\phi}|^{\text{ref}}$ ranging from 0.04 to 0.28 at $\phi_0 = 0.9$ and $f = 10$ Hz. The mean image corresponds to the time-averaged CH^* under steady conditions ($\phi = 0.9$). All images are normalized by the maximum intensity of this mean image.

regions, the absolute magnitude of ΔI increases with increasing $|\hat{\phi}|^{\text{ref}}$. This behavior is mainly attributed to the variation of the flame structure with the equivalence ratio. When the equivalence ratio exceeds 1.1, the chemiluminescence intensity near the flame root gradually decreases, while that in the upper region progressively increases. As the equivalence ratio further increases to 1.35, the flame completely detaches from the central bluff body and becomes stabilized approximately 2 cm downstream of the exit. In contrast, at the 180° phase, the spatial

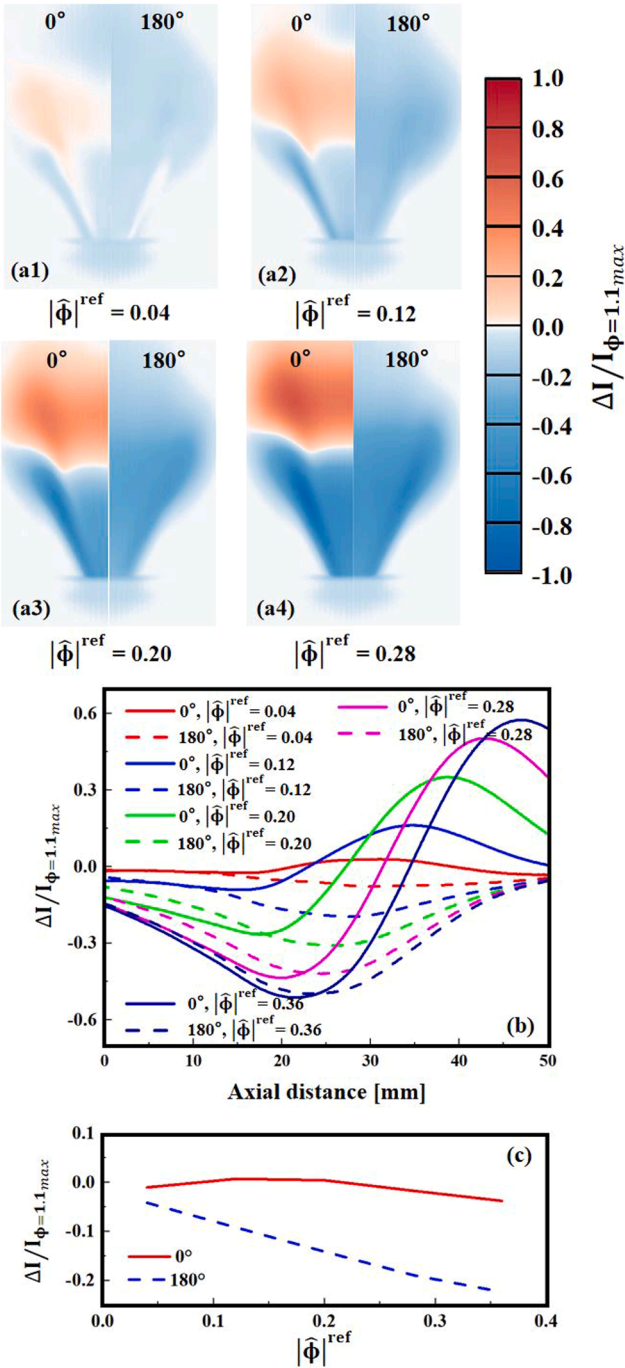


Fig. 6. (a) Phase-locked averaged CH* chemiluminescence images relative to the mean image at 0° and 180° phases, with their (b) radially averaged intensity along the axial direction, and (c) spatially averaged intensity for $|\hat{\phi}|^{\text{ref}}$ ranging from 0.04 to 0.36 at $\phi_0 = 1.1$ and $f = 10$ Hz. The mean image corresponds to the time-averaged CH* under steady conditions ($\phi = 1.1$). All images are normalized by the maximum intensity of this mean image.

distribution of ΔI exhibits a pronounced regional uniformity. And the overall chemiluminescence intensity across the entire flow field is uniformly lower than that of the steady flame at an equivalence ratio of 1.1.

The radially averaged ΔI profiles along the axial are shown in Fig. 6 (b). At the 0° phase, the interface between the positive and negative ΔI regions gradually shifts away from the burner exit. In the flame root, the chemiluminescence intensity initially changes rapidly with increasing $|\hat{\phi}|^{\text{ref}}$, followed by a slower variation. This transition occurs at $|\hat{\phi}|^{\text{ref}}$ of

approximately 0.2, where the instantaneous equivalence ratio corresponding to the maximum intensity reaches about 1.3, marking the critical point of flame morphological transition. A similar pattern appears in the upper region, although the shift from a rapid to a slower intensity increase is far weaker than that near the flame root. In contrast, at the 180° phase, the global CH* intensity varies nearly synchronously with $|\hat{\phi}|^{\text{ref}}$, and the overall spatial distribution shows no significant change. The globally averaged ΔI was then obtained, as shown in Fig. 6 (c). At the 0° phase, ΔI remains nearly constant over a wide range of $|\hat{\phi}|^{\text{ref}}$, and stays close to zero. While at the 180° phase, ΔI varies approximately linearly with $|\hat{\phi}|^{\text{ref}}$, which accounts for the observed phenomenon that the gain of the flame describing function is almost independent of the modulation amplitude.

4.2. Evaluation of the MISO model

Fig. 7(a–b) compare the flame describing functions for pure equivalence ratio oscillations at $\phi_0 = 0.9$ obtained from the dual-valve system and from the single-valve system combining with the MISO model. Over the examined range of $|\hat{\phi}|^{\text{ref}}$, the two methods produce highly consistent results in both gain and phase. As shown in Fig. 7(c), ϵ remains approximately below 10%, suggesting that the MISO model remains suitable for the experimental condition investigated. However, as illustrated in Fig. 4(a–b), the flame response under these conditions readily exhibits a certain extent of nonlinearity with increasing $|\hat{\phi}|^{\text{ref}}$, a scenario where the MISO model should be applied with caution. The suitability of the MISO model in the present work is likely attributed to the decreasing fractional contribution of velocity perturbations relative to equivalence ratio perturbation, namely $|G * (\hat{u} / \bar{u})| / |F^s * (\hat{\phi} / \bar{\phi})|$, with increasing equivalence ratio perturbation amplitude of $|\hat{\phi}|^{\text{ref}}$. For example, the velocity perturbation introduced by the single-valve system was measured to be approximately 9.8% of the bulk velocity, contributing to 51% and 6% of the chemiluminescence intensity fluctuation induced at amplitudes of $|\hat{\phi}|^{\text{ref}} = 0.02$ and 0.3, respectively, as shown in Fig. 7(c). Further sophisticated experiments with comparable contributions from decoupled velocity and equivalence ratio perturbations are required to

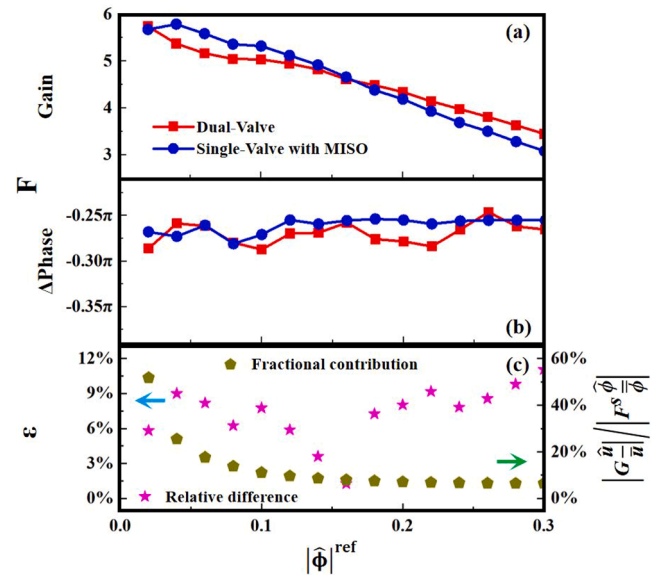


Fig. 7. The (a) gain and (b) phase of the describing function F obtained using the dual-valve system and the single-valve system in conjunction with the MISO model for the condition at $\phi_0 = 0.9$ and a fixed modulation frequency of 10 Hz (c) Relative difference, ϵ , determined from the difference between the results obtained using the MISO model and those measured directly. The fractional contribution of the velocity perturbation is defined as $|G * (\hat{u} / \bar{u})| / |F^s * (\hat{\phi} / \bar{\phi})|$.

fully explore the validity regime of the MISO model.

Then, the ξ fields were measured for the case of $|\hat{\phi}|^{\text{ref}} = 0.08$ (see Fig. 8(a)) to identify the origins of the relative difference ϵ . For consistency, all ξ fields were normalized by the maximum CH^* intensity of the steady flame at $\phi = 0.9$, yielding the normalized difference distributions. As shown in Fig. 8(a), at the 0° phase, ξ exhibits a pronounced vertical distribution, with positive values at the flame root and negative values at the flame tip. This indicates that the flame subjected to the single-valve modulation system has a slightly larger flame angle and a slightly lower flame height compared to the dual-valve modulation condition, presenting a morphology characterized by root expansion and tip contraction. The positive values in the root correspond to a relative increase in the chemiluminescence intensity, while the negative values at the tip indicate a relative decrease, reflecting a clearly non-uniform axial deviation between the two regions. In contrast, at the 180° phase, ξ primarily displays a lateral distribution, with negative values on the inner side and positive values on the outer side of the flame, indicating the radial variation in the flame's response to the coupled perturbations.

The observed differences can be further elucidated by considering the corresponding velocity perturbations (Figs. 8(b–c)). At the 0° phase, assuming a constant flame propagation speed, the reduced flow velocity leads to enhanced upstream propagation of the flame, thereby leading to an increase in the flame angle and a concomitant decrease in the flame height. Conversely, at the 180° phase, the increased flow velocity induces enhanced downstream propagation of the flame, resulting in a decrease in flame angle and an increase in flame height. These observations demonstrate that the geometric variations of the flame at different phases are the direct consequences of the combined effects of velocity and equivalence ratio oscillations, which simultaneously account for the spatial distribution differences shown in Fig. 8(a).

Moreover, the spatial distribution of the deviations introduced by using the MISO model is heavily influenced by the phase difference between velocity and equivalence ratio oscillations. When the phase of the velocity perturbation is shifted by π , the spatial distribution of ξ changes accordingly. Specifically, at the 0° phase with higher flow velocity, the flame angle decreases while the flame height increases, producing a spatial distribution opposite to that shown in Fig. 8(c). This indicates that the MISO model may locally underestimate or overestimate the chemiluminescence intensity variations in specific regions. Consequently, the deviations introduced by using the MISO model depend not only on the amplitude but also on the relative phase between the two perturbations. Accounting for this, phase dependence is crucial when evaluating the accuracy of the flame describing function constructed from the MISO model.

5. Conclusion

The present study systematically investigated the flame response to equivalence ratio (ϕ) oscillations of varying amplitudes at a fixed a modulation frequency of 10 Hz. Based on this, the multi-input single-output (MISO) model was evaluated by comparing the flame response to the ϕ -modulation introduced by both single-valve and a dual-valve modulation system. The dual-valve system enabled pure ϕ -modulation without introducing velocity oscillations, while both ϕ -modulation and velocity oscillations are present in the single-valve system.

The results demonstrate that the gain of the flame describing function exhibits an amplitude dependence for $\phi_0 = 0.65$ and 0.9 , indicating a nonlinear response at higher amplitudes ($|\hat{\phi}|^{\text{ref}} > 0.02$), while the phase remains constant. These observed trends are accurately captured by the QSSA predictions when a third-order Taylor expansion is applied. For $\phi_0 = 1.1$, the gain is largely amplitude-independent and can be accurately predicted by the QSSA at low amplitudes ($|\hat{\phi}|^{\text{ref}} < 0.1$). As the amplitude increases further, the equivalence ratio oscillations encompass the inflection point of the steady-state curve and even enter the regime of abrupt flame morphological changes, the deviations produced

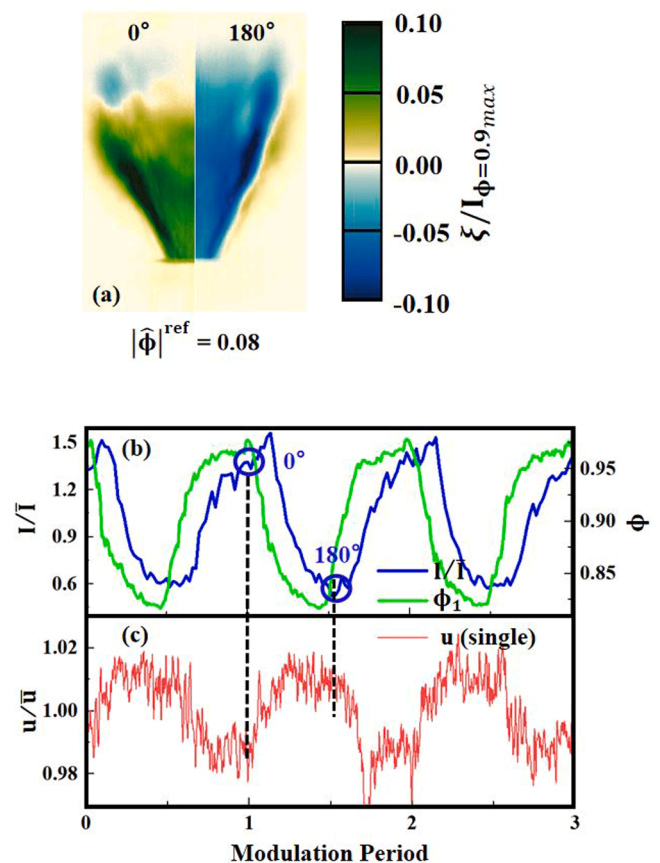


Fig. 8. (a) Spatial distribution of the normalized difference, ξ , at the 0° and 180° phases for $|\hat{\phi}|^{\text{ref}} = 0.08$, $\phi_0 = 0.9$ and $f = 10$ Hz. (b) Normalized CH^* chemiluminescence intensity and equivalence ratio oscillations in the time domain under single-valve modulation for $|\hat{\phi}|^{\text{ref}} = 0.08$, $\phi_0 = 0.9$ and $f = 10$ Hz. (c) Normalized velocity oscillation at Point 1 in the time domain under single-valve modulation.

by the QSSA predictions become even larger.

Under the assumption that velocity and ϕ oscillations are decoupled, the describing function obtained from the MISO model remains nearly accurate, even in nonlinear response regimes. This can be attributed to the decreasing fractional contribution of velocity perturbation relative to ϕ oscillations with increasing amplitude, $|\hat{\phi}|^{\text{ref}}$. Besides, the experimental data indicate that velocity perturbations alter flame angle and height, causing deviations in CH^* spatial distribution predicted by the MISO model. These spatial deviations are strongly dependent on the relative phase difference between the ϕ and velocity perturbations. Further experiments featuring comparable contributions from decoupled velocity and ϕ perturbations are recommended.

CRediT authorship contribution statement

Zhixin Zhao: Investigation, Formal analysis, Writing – original draft.
Weiyuan Liang: Investigation. **Dong Yang:** Writing – review & editing.
Bo Zhou: Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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